8.3 Activities: volume and angles

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1. Build the litre cube

You need: stiff card, a ruler, scissors, tape, a plastic bag or cling film, a measuring jug, rice or water.

Draw a net (*bouwplaat*) of an open cube with 10 cm sides: five squares of 10 × 10 cm in a cross shape. Fold and tape it. Line it with a plastic bag. Pour in exactly 1 litre of rice or water from a measuring jug. It fills the cube to the brim.

Talk about it: "So 10 cm × 10 cm × 10 cm is 1 000 cm^3, and it's also 1 litre. What does that make 1 cm^3?" (1 ml.)

2. Jug detective

You need: a measuring jug, a selection of containers (a mug, a glass, a yoghurt pot, a drinks bottle, a saucepan).

Before measuring, he guesses how much each container holds, in ml or l. Then fills it with water and pours it into the jug to check. Write guesses and real amounts side by side. Read the labels on drinks bottles and cartons too: 330 ml, 1,5 l, 25 cl.

Talk about it: "Which guess was closest? Did your guesses get better as you went?" (They nearly always do; that is what estimating practice looks like.)

3. Box volumes

You need: three or four boxes (shoebox, cereal box, tissue box), a ruler, sugar cubes or Lego bricks (optional).

First fill the bottom of a small box with sugar cubes to see "one layer". Count layers. Then measure each box in cm, round to whole cm, and calculate length × width × height. Turn the answer into litres (divide by 1 000). Which box holds the most? Is it the one that looks biggest?

Talk about it: "A cereal box is tall and thin. A shoebox is flat. Can they hold the same amount?"

4. Angle hunt

You need: a sheet of paper (its corner is a perfect right angle), a protractor.

Walk around the house and find: three right angles (*rechte hoeken*), two acute angles (*scherpe hoeken*), two obtuse angles (*stompe hoeken*) and one straight angle (*gestrekte hoek*). Use the paper corner to test: does the angle fit exactly, is it smaller, or bigger? Measure two of them with the protractor.

Good places: an open laptop, scissors, a clothes hanger, a door opened partway, the roof of a house across the street.

Talk about it: "Which was hardest to find? Why are there so many right angles in a house?"

5. Paper-folding angles

You need: a sheet of paper, a pencil.

Fold the corner of the paper in half: that is 45°. Fold a full sheet in half twice to make a right angle (90°) from a torn edge. Fold a circle in half (180°), quarters (90°), eighths (45°). Label each fold. Then check with the protractor.

Talk about it: “If you put two 45° angles side by side, what do you get? And four right angles?”

6. Clock angles

You need: an analogue clock or a drawn clock face with movable hands (a paper plate and a split pin work).

There are 12 numbers around 360°, so each number step is 30°. Set the hands to 3:00 (90°), 6:00 (180°), 1:00 (30°), 2:00 (60°), 4:00 (120°). He names the kind of angle each time and works out the degrees.

Talk about it: “At what times do the hands make a straight line?”

Conversation prompts for any day

- “This bottle says 1,5 l. How many 250 ml glasses is that?”
- “How many litres do you think the bath holds? How could we find out?”
- “Is the angle of that ramp acute or obtuse?”
- “When you turn all the way round on a swivel chair, how many degrees is that?”